Sean L. Snaider

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EDUCATION

Northeastern University **Boston, Massachusetts**

B.S. in Computer Science, Concentration in Software Engineering *Expected May 2028*

- **GPA:** 3.89/4.00, *Dean's List*

SKILLS

Languages: Python, Java, TypeScript, JavaScript, C, SQL
AI / LLM: Anthropic API, structured outputs, tool use, prompt engineering
AI Development Tools: Claude Code, GitHub Copilot, agentic coding workflows
Frontend: React, Next.js, Tailwind CSS, Zustand, Web Audio API
Backend & Data: FastAPI, Prisma, PostgreSQL, MongoDB, Pandas, NumPy, Polars
Tools & Infrastructure: Git, Docker, Linux/WSL, Vite, Vercel, GDB

EXPERIENCE

Vertex Pharmaceuticals **Boston, Massachusetts**

G&A Architecture and Strategy Co-op Jul 2026 - Dec 2026

- Building a Python tool to automate data migrations previously run by hand across multiple platforms and a legacy Microsoft Access workflow; projected by management to eliminate ~$100K in annual effort across business and IT teams.
- Designed the tool so non-technical staff can reconfigure migrations themselves without code changes, driving field mappings from declarative Excel configuration with per-field overrides rather than hardcoded transformation logic.

Student Government Association, Northeastern University **Boston, Massachusetts**

Software Engineer *Feb 2026 - Present*

- Replaced spreadsheet-based attendance and voting workflows with a full-stack web application now used by 50+ SGA members across 20+ officer meetings per semester.
- Moved vote validation from the browser to the server using Next.js API routes, Prisma, and schema-based request validation, closing a class of data-integrity bugs; deployed in production for SGA executive officer elections.

Khoury College of Computer Sciences, Northeastern University **Boston, Massachusetts**

TA for Program Design and Implementation 1 (Fall '25) & 2 (Spring '26) *Sep 2025 - Apr 2026*

- Supported 850+ students across two of Khoury's largest intro CS courses through weekly debugging hours covering OOP, data structures, recursion, and control flow.
- Coached students on effective use of Claude Code and GitHub Copilot in lab sessions and one-on-one, covering how to scope tasks, supply relevant context, and iterate when generated output missed intent.

Disrupt, Northeastern University **Boston, Massachusetts**

Software Engineer *Sep 2025 - May 2026*

- Contributed component refactors, UI fixes, and feature work to a production codebase alongside a team of 5+ developers.
- Taught 150+ students across 8+ recurring SQL and Python workshops, covering query optimization and joins.

PROJECTS

Guitar Learning Tool - *React, TypeScript, FastAPI, Anthropic API, Web Audio API - Github* *Jan 2026*

- Built browser-based chord detection with the Web Audio API, analyzing microphone input in the frequency domain to identify played chords in real time; wrapped in a practice-tracking app with a gamified scale-mastery progression and a 50-test async pytest suite.
- Integrated the Anthropic API to generate personalized chord progressions, augmenting the system prompt with structured practice history: a lightweight alternative to RAG.

Rubik's Cube Solver and Teaching Tool - *Python, React, TypeScript, FastAPI, MongoDB - Github* *Feb 2026*

- Reverse-engineered undocumented orientation requirements in Kociemba's algorithm to produce <20-move solutions in <1s; deployed as a full-stack app with CSS 3D cube rendering, now used as a competitive solve timer by 10+ peers.

Systems Programming - C, x86-64 Assembly, Linux *Fall 2025*

- Implemented a FUSE filesystem driver in C supporting standard file and directory operations over a disk image, with block allocation tracked by a hand-built bitmap.
- Built a dynamic memory allocator using a linked-list free list with first-fit allocation, and a Unix shell supporting piping and I/O redirection; used x86-64 assembly throughout to trace how compiled C maps to machine-level behavior.